Aryan Poonacha

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EXPERIENCE (*See my LinkedIn for more details).*

Tomex AI **March 2026 - Present**
Founding AI Engineer San Francisco

- Designed, built, led and owned various product surfaces, including the AI agentic chat application and cross-product integrations.
- Led AI and engineering infrastructure adoptions and optimizations - built dedicated independent sandbox environments for agents to test changes to code, dedicated in-house PR review bot customized to production codebase, CI/CD optimizations, and generally establishing and guiding engineering culture.

Lightfield (Formerly Tome) **February 2025 - March 2026**
Founding AI Product Engineer San Francisco

- Built & shipped core product features weekly (full-stack eng. in React, TS, GraphQL, Shad, Tailwind) with high ownership and project planning - drafting engineering sketches, splitting & planning tickets, managing scope, executing in collaborative CI/CD workflows.
- Owned product-customer forward-deployed feedback loop, working directly with design team to rapidly iterate features based on customer feedback, shipping PRs daily with no PMs. Worked directly with customers to investigate, resolve & document issues.
- Context engineering & tool calling construction, integrating custom tools, vector databases, RAG, and various domain-specific optimizations for Lightfield's custom agentic chat, enabling custom workflows and a truly integrated agent within software product.

Demandly (Formerly Dobbie) **May - December 2024**
AI Engineer Intern Remote

- Architected core agentic LLM flow, incorporating custom finetuned model and context construction tools via RAG & external APIs.

Origin AI **January - May 2024**
Product Intern Durham, N.C

- Interviewed customers, analyzed company data to compare & recommend specific go-to-market strategies for their latest products.

Duke University Computer Science Department **August 2022 - December 2024**
Teaching Assistant For CS 307 & CS 308 - Advanced Software Design & Implementation Durham, N.C

- Ran office hours & mentored 40+ students to work in teams on large software (ML and other) projects (Python, Java, Maven, git).
- Conducted 200+ code reviews, graded assignments. 100% pass rate and favorable performance reviews - rehired every semester.

Rhodes Information Initiative at Duke **May - July 2022**
Data+ Summer Research Assistant At Visualizing Durham Public Schools Project Durham, N.C

- Consulted with Durham Public School administrators, students and stakeholders to do feature selection and identify most relevant data points for users to iteratively develop data dashboard website in 2-week sprint cycles, achieving 3000+ monthly visits.

EDUCATION

Duke University Pratt School of Engineering **August 2023 - December 2024**
M.Eng. in Artificial Intelligence Durham, N.C

Duke University & Duke Kunshan University **August 2019 - May 2023**
Dual Degree B.Sc. in Data Science Durham, N.C and Kunshan, China

PROJECTS

ScriptScouts - Dynamic AI-Powered Webscraping API (Duke Engineering Consulting) **Spring 2024**

- Leveraged LLMs to monitor & maintain reliable webscraping scripts against dynamically changing content. Built custom self-adjusting API to generate web scraped JSON of dynamic drug prices - improving sensitive pricing data accuracy uptime by 2-3x.

Chatbot With Vector DB & Retrieval-Augmented Generation, *AI-590: Large Language Models* **Spring 2024**

- Deployed custom LLM chatbot w/ Retrieval-Augmented Generation system using vector database to answer domain-specific questions for any general given knowledge base. Implemented proof-of-concept for Duke AI Program & Video Game Wiki Chatbots.

Quantization Workbench, *AI-540: Deep Learning Applications Final Project* **Spring 2024**

- Created and deployed workbench tool to benchmark and compare different quantized versions of LLM models simultaneously.

Bass Connections - Analyzing Systemic Housing Inequality, *Under Dr. Thomas DeFrantz* **Fall 2021**

- Developed publicly showcased art gallery data display fluid simulation visualization model of racial housing inequality (Tensorflow).

HealthCarolina - Interactive Portal For North Carolina Health Info & Statistics, *Databases Final Project* **Fall 2021**

- Achieved 300+ monthly visits to interactive N.C map website highlighting health statistics based on geographical data modeling (GIS, numpy). Worked with a team of 4 to iterate features and optimize site for scale with load testing (Flask, Postgresql, MySQL).

AWARDS

- *GeorgiaTech's Robotech Hackathon 2022*: Best Overall Hack, Second Best Design, Most Accessible Hack. 3000$ in prizes and 5000$ in startup seed funding. National Finalist, *Alibaba GET Global Startup Challenge*.